

Quality Inspection Report QIR-2024-0871

Dimensional / Lot Acceptance Inspection | MANDREL, COMPOSITE, FILAMENT WOUND (incoming inspection)

Report No. / Date	QIR-2024-0871 / 2024-10-03
Part Number	ACT-FP-5500-01
Part Description	MANDREL, COMPOSITE, FILAMENT WOUND (incoming inspection)
Work Order / PO	Incoming / PO-87119 (Gulf Coast Composites LLC)
Lot / Heat-Batch	GCC-24-118 / Resin batch RB-24-0552
Lot Qty / Sample Qty	400 / 20 (per ACT-QP-08)
Serial Numbers	Lot sample
Material	E-glass / epoxy filament wound, 55 deg winding angle
Inspected By	D. Whitfield
Reviewed / Approved By	S. Nguyen
Applicable Specs	API 11D1 / API 19V (as applicable), NACE MR0175/ISO 15156, ASTM E709, API 5B

Dimensional & Attribute Results

Actuals are the worst-case (closest-to-limit) reading across the sample unless noted. Instruments are calibrated per ACT-CAL-01; calibration records on file.

Char.	Feature / Characteristic	Nominal	Tol +	Tol -	Actual (worst)	Instrument	A/R
D1	Mandrel OD	3.940	+0	--0.010	3.950	Micrometer M-215	ACC
D2	Bore ID	1.900	+0.010	-0	1.903	Bore gauge B-110	ACC
D3	Wall thickness (min)	1.000	+0.030	-0	1.014	UT thickness UT-7	ACC
D4	Length	14.5	+0.0	--0.0	14.5	Height gauge H-14	ACC
D5	Barcol hardness	62.0	+8.0	--4.0	67.6	Barcol tester BC-1	ACC
D6	Glass content (% wt, burn-off)	72.0	+3.0	--3.0	75.0	Lab furnace per ASTM D2584	ACC
D7	Compressive strength sample (ksi)	55.0	+99.0	--5.0	89.8	Lab press per ASTM D695	ACC

Result: **ACCEPT**

Measuring & Test Equipment Used

Instrument	Cal. Record	Cal. Due	Status
Micrometer M-215	CAL-8860	2026-07-14	In calibration at time of use
Bore gauge B-110	CAL-8367	2025-11-15	In calibration at time of use
UT thickness UT-7	CAL-8065	2026-04-16	In calibration at time of use
Height gauge H-14	CAL-9507	2026-11-27	In calibration at time of use
Barcol tester BC-1	CAL-0148	2025-10-06	In calibration at time of use
Lab furnace per ASTM D2584	CAL-7887	2026-07-20	In calibration at time of use

Instrument	Cal. Record	Cal. Due	Status
Lab press per ASTM D695	CAL-2687	2025-08-22	In calibration at time of use

Sampling & Methods

Sampling per ACT-QP-08 (ANSI/ASQ Z1.4 basis). Dimensional measurements at 68-72 degF after minimum 4 hr thermal soak. Surface finish per ASME B46.1. Hardness per ASTM E18 (HRC/HRB) or ASTM E140 conversion where noted. Threads gauged per API 5B with certified working gauges. Magnetic particle inspection, where applied, per ASTM E709 using wet fluorescent method, 100-125 V/in. field.

Inspector Notes

Lot accepted. Resin cure verified by DSC (Tg = 152 degC, min 145). Certificates of conformance on file. One mandrel per 200 destructively tested - burst 14,800 psi (min 12,500).

Sign-off

Role	Name	Signature	Date
QC Inspector (Level II)	D. Whitfield	/signed electronically/	2024-10-03
QA Manager	S. Nguyen	/signed electronically/	2024-10-03